Name ____ Date ___________ Period _____

Solving Absolute Value Linear Equations - Practice

Algebra II Honors | Unit 2 | Lesson 5 Practice | TEK A2.6.E

Objective: I can solve harder absolute value linear equations involving fractions and distribution, and justify the two-case split from the piecewise definition.

Section 1 — Recall

1. Solve $|x| = 21$, then explain in one sentence why there are two solutions.

x = ________ or x = ___________ ____________________________

2. Solve $|4x + 3| = 17$.

x = ________ or x = ________

Section 2 — Apply

3. Solve $5|x - 2| - 3 = 22$.

x = ______ or x = ________

4. Solve $|(x + 6)/2| = 7$.

x = ______ or x = ___________

Section 3 — Challenge

5. Solve $|4(x - 3)| + 5 = 29$.

x = ______ or x = ________

6. Solve $9 - 2|x + 4| = 1$.

x = ______ or x = ________

Section 4 — Explain It

7. Using the piecewise definition of absolute value, explain WHY an equation like $|A| = k$ (for $k > 0$) always splits into exactly two cases: $A = k$ or $A = -k$.

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